



Australian Government
Australian Research Council

PROGRESS REPORT ON ARC FUNDED RESEARCH - 2010

(Discovery Projects)

Purpose

The ARC requests a progress report from each Project Leader to assess progress and assist the researcher in cases where this progress has not been satisfactory or where difficulties have been encountered.

Instructions

Progress reports for *Discovery Projects* are required to be submitted to the ARC by 28 February 2011.

The Project Leader must note:

- A Progress Report for each ARC funded research project must be completed by the **Project Leader** named on the project.
- Please expand tables as appropriate.
- While there is no limit to the amount of text that can be provided, the ARC is generally seeking **1 to 2 paragraphs of information** where a textual response is required.
- Once complete, the Project Leader should submit the **electronic copy** of the progress report to the Research Office of the Administering Organisation. Paper copies are not required.
- The completed progress report should be saved as a rich text format (RTF) or a PDF document. The file name for the progress report should conform to the following naming convention:
[ProjectID][CI1Surname]_[year].rtf or pdf
(For example DP012345678Smith_2010.rtf)
- The saved RTF or PDF file should be attached to an **email** and sent to the Research Office of the Administering Organisation.

Research Administrators must note:

- Progress Reports should be submitted to the ARC in **electronic (RTF or PDF) format only**:
- The electronic copy (RTF or PDF) must be emailed to the ARC at progress.reports@arc.gov.au.
- Please ensure the file name for the electronic version conforms to the naming convention detailed above (files inappropriately named will be returned to the Research Office for action).

PROGRESS REPORT ON ARC FUNDED RESEARCH FOR *DISCOVERY PROJECTS* ACTIVE IN THE YEAR 2010

1. PROJECT IDENTIFICATION

1.1 Project ID

DP1094883

1.2 Administering Organisation

Macquarie University

1.3 Project start date and total amount of ARC funding received for this reporting period

1 January 2010

\$120075

1.4 Project title

Applicable categorical structures

1.5 Project Leader

Name	Emeritus Professor RH Street
Email address	ross.street@mq.edu.au

2. PROJECT DESCRIPTION AND OBJECTIVES

2.1 100 word Project summary [as indicated in the original proposal]

Sets with structure are to modern mathematics what numbers were to the ancients. Such structures can often be modelled in monoidal categories vastly different from the category of sets. In algebraic cases, there are free structures formed from symbols with no extra relations beyond those required to express the structure. Often, also, examples of such models in categories whose ingredients are geometric concepts (such as tangles on strings or cobordisms) have freeness properties that involve moving out of that base geometric category. This kind of freeness has allowed the construction of important invariants of geometric figures. Our project will study both theoretical and applied aspects of this encompassing schema.

3. PROJECT OVER DURATION OF FUNDING

The information you provide in this section is confidential and will not be used for media or promotion of research.

3.1 Have there been changes to or problems with the project?

Yes

(yes/no)

This could include changes to the research project resulting from ARC funding at a lower level than requested. Note that some changes require ARC approval to be requested as a Variation to the Funding Contract/Agreement. Such a Variation must be forwarded to the ARC through the Administering Organisation's Research Office. Variations to Contract/Agreement forms are available on the ARC website at www.arc.gov.au

If yes, give details

A change of candidate led to a late appointment of research associate.

3.2 What are your achievements and any research findings to date?

Please provide 1 to 2 paragraphs only

In this first year of the grant, much work has centred on the postgraduate students whose topics are part of the project: David Oury, Micah McCurdy, Matt Roberts, Thomas Booker and, to some extent, Kamel Kachour. Categorical structures in computer science were developed by CI Johnson and colleagues. Johnson chaired AMAST (Algebraic Methodology and Software Technology) held in Quebec City in

June and lectured at the University of Milan in April. The main excitement in the work is the newly opened avenues of research presented in papers [4] and [7] which led to Johnson's invitation to Dagstuhl. CI Street was on the Program Committee for the Category Theory 2010 Conference in Genoa (Italy) where CIs Lack and Verity, and students Booker and McCurdy contributed talks.

Our progress can be measured by our research documents:

1. Ignacio Lopez-Franco, Ross Street and Richard Wood, Duals invert, *Applied Categorical Structures* **19** (February 2011) 321–362.
2. Michael Johnson and Robert Rosebrugh, Ontology engineering, universal algebra and category theory, Invited book chapter in *Theory and Applications of Ontologies* volume 2: The Information-science Stance (edited by Roberto Poli, Achilles Kameas and Michael Healy; Springer-Verlag, 2010) Chapter 24.
3. Ross Street, Maria Manuel Clementino and Samson Abramsky (Editors of Special Issues devoted to CT2007 at Carvoeiro, Algarve, Portugal): "Toposes and Functor Categories" *Applied Categorical Structures* **18(2)** (April 2010) and "Enriched and Higher Categories" *Applied Categorical Structures* **18(3)** (June 2010).
4. Michael Johnson, Robert Rosebrugh and Richard Wood, Algebras and Update Strategies, *Journal of Universal Computer Science* **16(6)** (2010) 729-748.
5. Michael Johnson, and Stefano Kasangian, A Relational Model of Incomplete Data without `{tt NULL}s`, *Conferences in Research and Practice in Information Technology* **109** (2010) 89-94.
6. Gabriella Böhm, Stephen Lack, and Ross Street, On the 2-category of weak distributive laws, *Communications in Algebra* (accepted 14 Nov 2010).
7. Michael Johnson, Robert Rosebrugh and Richard Wood, Lenses, Fibrations, and Universal Translations (submitted).

3.3 What are your research milestones for the coming year?

Please provide 1 to 2 paragraphs only

(Please note that in your next report, you may be asked to report progress against these plans and objectives.)

1. Four lectures entitled "Monoidal categories in, and linking, geometry and algebra" will be delivered In May on the occasion of Street's award of the *Chaire de la Vallée Poussin 2011* at the ancient Catholic University of Louvain, Belgium.
2. Richard Garner was to be employed under this grant however was awarded his own Australian Research Fellowship with our research group. He is an organiser for, and he and Verity will be attending, a workshop at the prestigious Mathematisches Forschungsinstitut Oberwolfach (Feb 27- Mar 5). The workshop is entitled "The Homotopy Interpretation of Constructive Type Theory". Other participants will include Andre Joyal, Per Martin Lof and Vladimir Voevodsky. This is extremely relevant to the homotopy theory / type theory material in our project and shows how significant this aspect is becoming.
3. Lack is on the scientific committee for Category Theory 2011 (Vancouver) and will participate; he will also visit the Catholic University of Louvain in September, and attend the *Workshop on Categorical Algebra 2: Recent developments and future perspectives* in Gargnano, Italy.
4. Lack and Street plan to complete another joint paper with Gabriella Böhm.

5. Lack will complete papers on “Twisting structures” with Kathryn Hess, on “An embedding theorem for adhesive categories”, and on “Exactness properties involving finite limits” with Richard Garner. His work on “Homotopy locally presentable categories” with Jiri Rosicky will continue.
6. Verity will continue to develop his theory of “Higher Derivateurs”, whose purpose is to provide an abstract framework for describing the (co)completeness properties of categories of quasi-category-like structures. Higher derivateurs take their motivation from Alex Heller’s “homotopy theories” and Alexander Grothendieck’s “derivateurs”. The work grew directly out of feedback from Verity’s 2010 Morgan-Phoa Workshop presentation concerning a previous grant project. A complete set of axioms for the abstract theory of flexible limits and colimits in quasi-category theories will be obtained and extended to address the process of re-constructing certain kinds of higher homotopical information from 2-dimensional structures.
7. Street, Verity with Oury will continue investigating weak 2-categories as Theta-sets.
8. Verity will continue the theory of weak complicial sets – including extensions of certain coherence/strictification results, some work with Garner towards relating weak complicial sets to a more algebraic/operadic weak category variant, and an account of certain weak complicial sets of generalised cobordisms.
9. The above papers [4], [5], [7] have resulted in ongoing work for Johnson in 2011. Paper [4] was unexpected, and surprisingly integrative by explaining and connecting work going back to the 1980s with current work on bidirectional transformations, while [7] broke new ground in bidirectional transformations leading to the debunking of a “folk-theorem”. Further papers will be prepared on this.
10. With Dusko Pavlovic, Johnson will edit the volume “Algebraic Methodology and Software Technology” as *Lecture Notes in Computer Science* **6486** (2011).
11. The following 4 students will submit their PhD theses: David Oury (“Duality for Joyal’s category Theta and homotopy concepts for Theta-sets”, supervisors Street and Verity); Micah McCurdy (“Cyclic star-autonomous categories and the Tannaka adjunction via graphical methods”, supervisor Verity); Matt Roberts (“Compiling generics in functional programming languages”, supervisor Verity, associate supervisor Sloane); Thomas Booker (supervisor Street).
12. Micah McCurdy will join us as a research associate on the project.

4. ATTACHMENTS

Is additional material attached?

(yes/no)

5. CERTIFICATIONS

5.1 Certification by the Project Leader

I certify that:

All details in this report are true and complete and that this is an accurate Progress Report for the period covered.

Project Leader

Name: Emeritus Professor Ross Street	Date: 11/02/2011
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5.2 Certification by Deputy Vice-Chancellor (Research)

Deputy-Vice Chancellor (Research) (or equivalent or nominee) - comments

I certify, to the best of my knowledge, that progress on this Project has been satisfactory and/or problems have been identified in this Progress Report, and I recommend continued funding.

Deputy Vice-Chancellor (Research) or equivalent or nominee

Name:	Date:
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Please note: The electronic submission of this Progress Report to the ARC by the Research Office of the Administering Organisation is verification that the Administering Organisation has approved the Progress Report content.

An electronic signature is not required.

Thank you for submitting your Progress Report.

The ARC may contact you if clarification or further information is required to determine whether progress has been satisfactory over the period covered in the Report.

Information on this form is collected in order to determine whether the research project funded by the ARC has reached satisfactory outcomes for the year and for post award reporting.

Researchers should note that if the ARC is not satisfied with the progress of the Project, further payment of funds will not be made until satisfactory progress has been made on the Project. If satisfactory progress is still not achieved, the Funding will be terminated and all outstanding monies recovered by the ARC. Unsatisfactory progress on the Project will be noted against any further proposals under any ARC scheme submitted by, or on behalf of, the Chief Investigator or Fellow and will be taken into account in the assessment of those proposals.